

Discrete Ramsey Model *

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*For more discussions, please refer to Barro and Sala-i-Martin [2], and Acemoglu [1].

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1 Introduction

We have talked about Solow growth¹ model before. In Solow growth model, we assume that the households save a constant fraction $s \in (0, 1)$ of their income. Here, we relax this assumption. We assume that each household wants to optimize its utility of the whole life. So the savings will be determined endogenously. This is also called Ramsey-Cass-Koopmans (RCK) model.

For the facts about long-run economic growth, please refer to Professor Zhang's lecture notes in "Elearning".

2 Social Planner's Economy

2.1 The Economy

- The economy is an isolated island. There is a social planner.
- There is only one good, which can either be consumed after production or saved for investment at next period.
- Population keeps constant at 1. Labor supply $n_t \in [0, 1]$.
- Suppose that existing capital depreciates at a fixed rate $\delta \in [0, 1]$. So the accumulation of capital follows

$$K_{t+1} = (1 - \delta)K_t + i_t \quad (1)$$

2.2 Production

- The production function for final good is written as

$$Y_t = F(K_t, N_t) \quad (2)$$

where Y_t is the total output of the final good, K_t is the capital stock, $N_t = \gamma^t n_t$ is the augmented labor input. $A_t = \gamma^t$ denotes the technology. We also assume that $F : R_+^3 \rightarrow R_+$ is twice differential in K and N .

- F satisfies "neoclassical" properties:

1. Constant returns to scale in K and n ,

$$F(\mu K_t, \gamma_t \mu n_t) = \mu F(K_t, \gamma^t n_t)$$

2. Positive and diminishing marginal products:

$$F_K(K, An) \equiv \frac{\partial F(K, An)}{\partial K} > 0, F_n(K, An) \equiv \frac{\partial F(K, An)}{\partial n} > 0$$

$$F_{KK}(K, An) \equiv \frac{\partial^2 F(K, An)}{\partial K^2} < 0, F_{nn}(K, An) \equiv \frac{\partial^2 F(K, An)}{\partial n^2} < 0$$

¹This is called "exogenous growth" in Ljungqvist and Sargent [3].

3. Inada conditions:

$$\lim_{K \rightarrow 0} F_K(K, An) = \infty \text{ and } \lim_{K \rightarrow \infty} F_K(K, An) = 0 \text{ for all } L > 0 \text{ and all } A.$$

$$\lim_{n \rightarrow 0} F_L(K, An) = \infty \text{ and } \lim_{n \rightarrow \infty} F_L(K, An) = 0 \text{ for all } K > 0 \text{ and all } A.$$

- Using Euler's Theorem², we can have the following result from the above assumptions:

$$Y_t = F_n(K_t, A_t n_t) n_t + F_K(K_t, A_t n_t) K_t$$

2.3 Social Planner's Problem

Social planner tries to maximize the utility of whole life by choosing the optimal path of $\{c_t\}_{t=0}^{\infty}$ and $\{n_t\}_{t=0}^{\infty}$. Thus the problem of social planner would be

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \\ \text{s.t.} \quad & c_t + K_{t+1} \leq F(K_t, \gamma^t n_t) + (1 - \delta) K_t, \\ & n_t \in [0, 1], \\ & K_0 > 0 \text{ is given.} \end{aligned} \tag{3}$$

THEOREM 11 *Balanced growth path is possible as a solution to the social planner's problem if and only if*

$$u(c, n) = c^{1-\sigma} v(1 - n)$$

where $v(\cdot)$ is a function of leisure.

Here, balanced growth path means a situation where all endogenous variables grow at constant (but possibly different) rates.

2.4 Model Transformation

Given the theorem above, we know there may be no balanced growth for general case. We come across a big problem for we to find the balanced growth path here. Recall the meaning of balanced growth path is

$$\begin{aligned} Y_t &= Y_0 g_y^t \\ c_t &= c_0 g_c^t \\ K_t &= K_0 g_K^t \\ i_t &= i_0 g_i^t \end{aligned}$$

Combine these equations with the capital accumulation law (1), we know

$$g_K = g_i.$$

²Refer to the Mathematical Appendix M.B of Mas-Collel, Whinston and Green[4].

Similarly from the resource constraint $c_t + i_t = Y_t$, we have

$$g_y = g_c = g_i = g_K$$

. Our production function $F(\cdot, \cdot)$ is constant return to scale. We have

$$F(K_t, \gamma^t n_t) = Y_t = Y_0 g_y^t = F(K_0, n_0) g_y^t = F(K_t, g_K^t n_0)$$

Thus,

$$g_y = g_c = g_i = g_K = \gamma$$

. So our method is to divide each variable by its growth rate:

$$\begin{aligned}\hat{c}_t &= \frac{c_t}{\gamma^t} \\ \hat{i}_t &= \frac{i_t}{\gamma^t} \\ \hat{K}_t &= \frac{K_t}{\gamma^t} \\ \hat{Y}_t &= \frac{Y_t}{\gamma^t}\end{aligned}$$

And the problem of the social planner has been transformed into

$$\begin{aligned}\max \quad & \sum_{t=0}^{\infty} \beta^t u(\gamma^t \hat{c}_t, n_t) \\ \text{s.t.} \quad & \hat{c}_t + \gamma \hat{K}_{t+1} \leq F(\hat{K}_t, n_t) + (1 - \delta) \hat{K}_t, \\ & n_t \in [0, 1], \\ & K_0 > 0 \text{ is given.}\end{aligned} \tag{4}$$

3 Market Economy

3.1 Households

- Households live an infinite amount of time.
- Each person in a household supplies one unit of labor inelastically with the wage rate w_t .
- Household will accumulate wealth a_t , and lend them to the firms with the return rate R_t .
- The wealth depreciates³ at rate $\delta \in (0, 1)$.
- Each household's period utility is given by $u(c_t, n_t)$.
- Household tries to maximize the utility of whole life by choosing the optimal path of $\{c_t\}_{t=0}^{\infty}$ and $\{n_t\}_{t=0}^{\infty}$.

The problem of each household would be

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \\ \text{s.t.} \quad & c_t + a_{t+1} \leq w_t n_t + R_t a_t + (1 - \delta)a_t, \\ & n_t \in [0, 1], \\ & K_0 > 0 \text{ is given.} \end{aligned} \tag{5}$$

3.2 Firms

- Firms employ labor and rent capital in competitive labor and capital markets to produce the final good.
- The technology for each firm j is $Y_t^j = F(K_t^j, \gamma^t n_t^j)$.
- The firms seek to maximize profits. The FOCs for an interior solution require

$$\begin{aligned} F_K(K_t^j, \gamma^t n_t^j) &= R_t \\ F_n(K_t^j, \gamma^t n_t^j) &= w_t \end{aligned}$$

3.3 General Equilibrium

- Market Clearing:

$$\int_j K_t^j dj = K_t \text{ and } \int_j n_t^j dj = n_t$$

³A different way to understand this setting is that household accumulate capital, and the firms rent the capital from households.

- Given the market structure, we know

$$R_t^j = R_t^i, \text{ and } w_t^j = w_t^i$$

Thus all the firms have the same $\frac{K}{L}$ ratio. So the production function of the whole economy can be written as:

$$Y_t = F(K_t, \gamma^t n_t) \quad (6)$$

And the problem of the household can be re-written as

$$\begin{aligned} \max \quad & \sum_{t=0}^{\infty} \beta^t u(c_t, n_t) \\ \text{s.t.} \quad & c_t + K_{t+1} \leq F(K_t, \gamma^t n_t) + (1 - \delta)K_t, \\ & n_t \in [0, 1], \\ & K_0 > 0 \text{ is given.} \end{aligned} \quad (7)$$

References

- [1] Darron Acemoglu, 2009, *Introduction to Modern Economic Growth*, Princeton University Press.
- [2] Robert Barro, and Xavier Sala-i-Martin, 2004, *Economic Growth*, 2nd Edition, The MIT Press.
- [3] Lars Ljungqvist, and Thomas J. Sargent, 2004, *Recursive Macroeconomic Theory*, 2nd Edition, The MIT Press.
- [4] Andreu Mas-Collel, Michael Whinston, and Jerry R. Green, 1995, *Microeconomic Theory*, Oxford University Press.